



**SERANGOON JUNIOR COLLEGE
2008 JC2 PRELIMINARY EXAMINATIONS**

MATHEMATICS

Higher 2

9740/Paper 2

Monday

25 August 2008

Additional materials: Writing paper

List of Formulae (MF15)

TIME : 3 hours

READ THESE INSTRUCTIONS FIRST

Write your name and class on the cover page and on all the work you hand in.

Write in dark or black pen on both sides of the paper.

You may use a soft pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use a graphic calculator.

Unsupported answers from a graphic calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphic calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [] at the end of each question or part question.

At the end of the examination, fasten all your work securely together.

This question paper consists of 7 printed pages and 1 blank page.

[Turn Over

Section A: Pure Mathematics [40 marks]

- 1** **(i)** Newton's Law of Cooling states that the rate at which an object cools is proportional to the difference in temperature between it and its surroundings.

If y is the temperature of an object (in degree Celsius) at time t (in hours) and θ is the temperature of its surroundings (in degree Celsius) which remains constant, form a differential equation relating y and t . **[2]**

- (ii)** Professor Mandrake was found murdered in the library. The ambient temperature is a constant 26°Celsius . The temperature of the body was 30°Celsius at 7 am and it was 28°Celsius at 9 am. Assuming his body temperature was known to be 37°Celsius at the time of his death. When was he murdered?
Give your answer to the nearest minute. **[7]**

- 2** A sequence $u_1, u_2, u_3, \dots$ is such that $u_1 = \frac{1}{4}$ and $u_{n+1} = u_n - \frac{6}{(3n+5)(3n+8)}$, for $n \geq 1$.

- (i)** Find the exact values of u_2, u_3, u_4 and u_5 . **[2]**

- (ii)** Show that the conjecture for u_n in terms of n is given by $\frac{2}{3n+5}$. **[1]**

- (iii)** Prove the conjecture using mathematical induction. **[4]**

- (iv)** Hence find $\sum_{n=1}^N \frac{3}{(3n+5)(3n+8)}$ in terms of N and state its value as $N \rightarrow \infty$. **[3]**

3 (i) Show that $\frac{1+e^{i\theta}}{1-e^{i\theta}}$ can be reduced to $i \cot\left(\frac{\theta}{2}\right)$. [2]

(ii) Find all the roots of the equation $z^8 = \frac{1}{\sqrt{2}}(1-i)$, giving each root in the form $re^{i\theta}$ where $r > 0$ and $-\pi < \theta \leq \pi$. [5]

(iii) Hence deduce all the roots of $\left(\frac{w-1}{w+1}\right)^8 = \frac{1}{\sqrt{2}}(1-i)$ [3]

4 Show that

$$\int_a^b \frac{8x+1}{4+3x^2} dx = \frac{4}{3} \ln \frac{4+3b^2}{4+3a^2} + \frac{1}{2\sqrt{3}} \left[\tan^{-1}\left(\frac{\sqrt{3}b}{2}\right) - \tan^{-1}\left(\frac{\sqrt{3}a}{2}\right) \right] \quad [4]$$

A region is bounded by the following:

$$y = \frac{8x+1}{4+3x^2}; \quad y = \frac{57}{32}x; \quad y = 0 \quad \text{and} \quad x = 2$$

(i) Find the exact value of the area of this region. [4]

(ii) Find the volume generated by this region when it is rotated 2π radians about the x -axis, giving your answer correct to 3 decimal places. [3]

Section B: Statistics [60 marks]

- 5 A group of 7 boys and 5 girls are going to a concert. Given that the 5 girls want to sit together, find the number of ways to seat the 12 students
- (i) in a single row, [1]
 - (ii) in 2 rows of 6 each, [3]
 - (iii) at a round table if they are going for dinner after the concert. [2]
- 6 A particular neighbourhood teleshop, opens six days a week, finds that its sales of new and used hand phones both follow a Poisson distribution independently with means 5.4 and 9 per week respectively.
- (i) Find the average number of hand phones sold in a day. [1]
 - (ii) A day with more than the average number of hand phones sold in a day is described as a good day. Find the probability of having any one day described as good. [2]
 - (iii) In a particular day, it is given that three hand phones were sold. Find the probability that one more new hand phone was sold than the used hand phone on this particular day. [3]
- 7 Liquid detergent is sold in bottles of two sizes. The mass of each small bottle may be regarded as a normal variable with mean 505 g and standard deviation 10 g while the mass of each large bottle may be regarded as another independent normal variable with mean 1005 g and standard deviation 20 g.
- (i) Find the probability that the mass of one randomly chosen large bottle is less than 1000 g. [1]
 - (ii) Find the probability that the mass of one randomly chosen large bottle exceeds the total mass of two randomly chosen small bottles. [2]
 - (iii) Three large bottles and two small bottles are chosen at random. Find the probability that the average mass of all the five bottles chosen lies between 800 g and 815 g. [4]

8 (a) What is the main difference between the stratified sampling and quota sampling methods? [1]

(b) (i) Briefly describe how stratified sampling is being carried out and give an advantage of this sampling method. [3]

(ii) Briefly describe how quota sampling is being carried out and give an advantage of this sampling method. [3]

9 Peaches produced by a fruit grower are packed into boxes to be sold every day. It is known that about 1% of these peaches are infected. A check is carried out on a **large number, n** , of boxes by taking a random sample of 10 peaches from each box and examining them for infection.

Denoting the number of infected peaches in each sample of 10 peaches by X , and letting T be the sum of the number of infected peaches in the n samples, find the smallest number of n for $P(T < 0.145n) > 0.9$. [6]

State the approximate value of $P(T > 0.145n)$ as n becomes very large, justifying your answer. [1]

- 10** An engineer collected data on how the temperature, T °C, generated in the shoulder of a car tyre varies with its speed, V km/h, when the car is driven under specified conditions.

He found that the two variables follow a linear model with regression line of T on V as $t = 0.8667v + 26.5833$ with product-moment correlation coefficient, $r = 0.9939$.

As a backup, the engineer's assistant was also on site to record the temperature readings when the experiment was being carried out. The assistant submitted his observations as shown in the table below:

v	20	30	40	50	60	70	80	90
t	45	52	64	66	66	89	98	104

It came to the engineer's attention that his assistant's temperature readings contained an error. Having misplaced his original records, the engineer decided to work out the correct data. He is certain that the wrong t value (the corresponding v value remains correct) occurred in one of these three data points: (60, 66), (70, 89) or (80, 98).

- (i) Which one of the above three data pairs is most likely to contain the wrongly entered value of t ? Show clearly how you are able to identify the error. **[2]**
- (ii) Find the correct value of t recorded by the engineer giving your answer correct to 3 significance figures. **[3]**
- (iii) With the use of a suitable regression line, estimate the value of v given that $t=70$. Comment on the reliability of your estimation. **[3]**

- 11 (a)** A random sample of 100 wooden rods of a certain length is measured and the length x in metres is recorded. The results are summarised as follows:

$$\sum (x - 30) = -150, \quad \sum (x - \bar{x})^2 = 1485.$$

A test was carried out at the 2% significance level with the following hypotheses:

H_0 : the population mean length of the wooden rod is μ_0

H_1 : the population mean length of the wooden rod is $\neq \mu_0$

Given that H_0 is rejected in favour of H_1 , find the set of possible values of μ_0 . Assume that the distribution of the length of the wooden rod is normal. [5]

- (b)** In an industrial experiment to study the speed of performing a specific assignment, 10 workmen were chosen at random and the time (minutes) taken by a workmen to perform the assignment were taken as follow.

12.6 12.1 11.0 11.8 12.3 10.3 12.4 12.9 11.6 12.2

Perform an appropriate test at the 5% level of significance to check if the mean time taken to perform the task is no longer than 12 minutes. State one assumption that is used in the test. [4]

- 12** A bag contains 3 red balls and 3 black balls. Whenever a red ball is drawn it is replaced in the bag and an extra red ball is added to the bag. Whenever a black ball is drawn it is not replaced and no extra balls are added. 3 balls are drawn, one after another. Construct a probability tree showing this information. [2]

Find the probability that

- (i) all the balls are red, [1]
 (ii) at least one of the balls is red, [2]
 (iii) exactly two of the balls are red. [2]

Given that exactly 2 of the 3 balls drawn are red, find

- (iv) the probability that the first ball drawn is red. [3]

End of Paper

Blank Page